

PHASE 2 - CAPA INVESTIGATION RESULTS & ROOT CAUSE ANALYSIS

CAPA Investigation Results Assigned to:	Jacqueline Kim	CAPA Investigation Results Target Due Date:	03/31/2025	CAPA Investigation Results Completion Date:	3/26/2025
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CAPA Investigation Results: *(Describe the investigation and results that determined what cause (or will cause) the incident/problem to occur. Attach any supporting documents. If needed, complete and attach the SCAR Form, FM-0087)*

1. 100% of the DHRs for 2024 in Arena were reviewed (n=69)

a. 240 Training records reviewed

Training Issues	Identifier	Qty	%
Missing Training Records	Missing	16	6.67%
Training Performed >1yr before Manufacturing	Old	30	12.5%
Training Performed after Manufacturing	Late	30	12.5%
Total Issues in Training		76	32%

b. The investigation showed that late and missing training could be further broken down by project

Late and Missing Records by Project 2024

Project	Percentage
ALT	21.7%
UPL	30.4%
Arga	21.7%
Other	26.1%

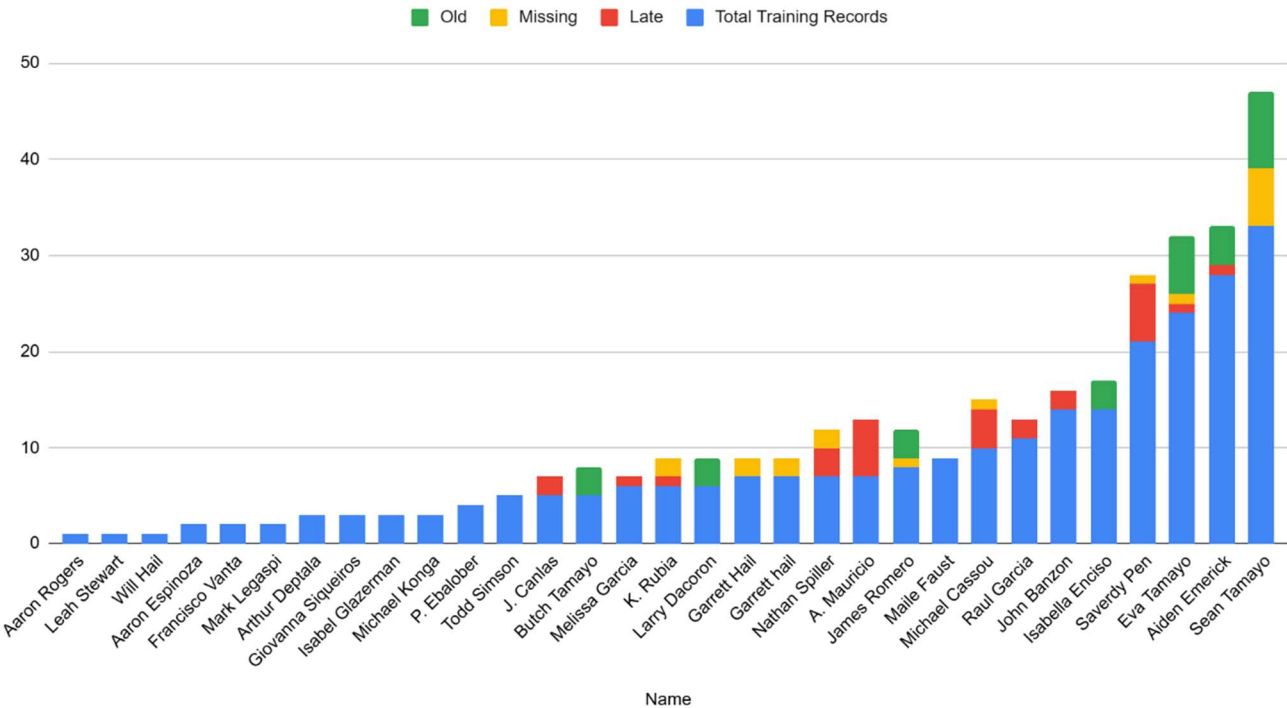
ALT and UPL projects were overseen by engineering. Production claims to have training records, but records were not available in Arena. Arga oversees their own training and performed a group training in Feb 2025 when training records were discovered to be missing from Arena.

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Projects for clients that required engineering and DV support lacked design transfer, where training would usually be evaluated and assigned. There was a limited number of manufacturing protocols that referenced training requirements; however, it was discovered that a portion of those MPs had a note for training to be listed in a separate section, but the section did not have the required content for training. Test Protocols (TPs) contained training requirements but lacked specificity to documents.

c. Training record assessment

Total Training Completed in 2024



i. 48.5% of individuals had 0 training issues; 32.5% had <50% training issues; and 19% had >50% training issues

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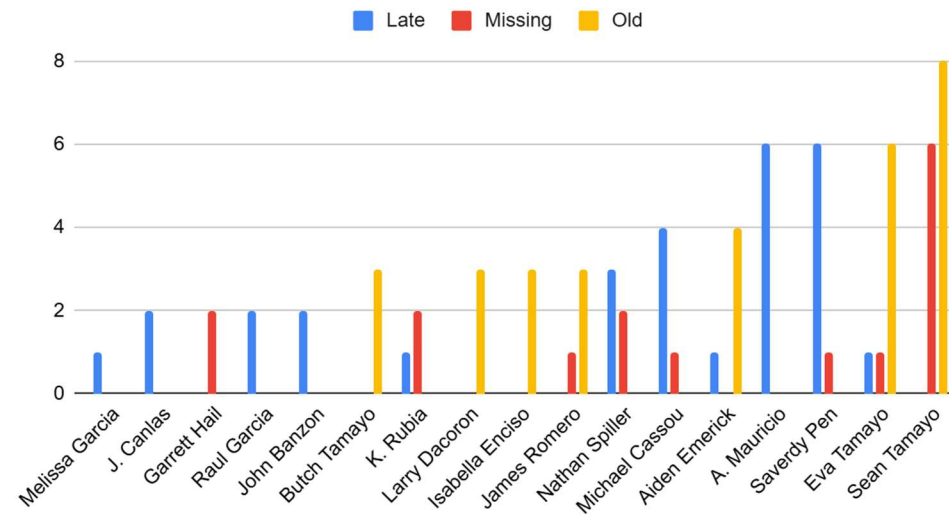
d. Training issue assessment (Removed ALT, Arga, and UPL)

Breakdown of Training Issues by Type



i. MPs and WIs were most likely to be late or missing

Training Issues per Individual



ii. The majority of individuals with late training are in short-term manufacturing only

iii. Individuals with the most old or missing training were long-term employees or employees with multiple responsibilities

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	2. 40% of the IQ/OQ/PQ Files for 2024 in Arena were sampled (n=35) a. 2 out of 35 training records were missing or not available for this review i. The 2 missing records were for IQ only ii. The records with missing training records were counter signed/reviewed by individuals trained to the appropriate protocol (QP-012) b. All OQ and PQ had appropriate training records stored within the report item in Arena 3. QPs were reviewed for all FTEs a. 5 of 18 FTEs were missing 1-2 training records for protocols or work instructions required to perform their current role i. 4 of the 5 missing training files were due to an increase in responsibility outside the original scope of their job description ii. 1 of the 5 missing training files was due to a clerical error on assigning training b. QA recommends increasing the scope of training for 8 FTEs to increase overall competency of QMS activities in Pathway 4. Observations from Investigation a. All training was completed after the release of the documents. No instances of training before release. b. Multiple training records exist for the same protocol and revision for some individuals c. Multiple instances of records not completely filled i. The title contains record trained to but not the training form ii. Signature, but no printed names d. Training records not all linked to individual training plans
Impact / Risk Assessment: <i>(Determine the impact/risk of the problem/incident or potential problem/incident on the product, processes, systems, documentation, etc. Determine the level of severity and probability of each risk using QP-004, Risk Assessment)</i>	For R&D and engineering builds, the risk is low, as the units are meant for proof of concept and data gathering. For DV/V&V units, the risk is low, as lack of training reduces confidence in traceability in manufacturing. If a manufacturing or in-process step is performed incorrectly, verification testing may be impacted. If the device lacks traceability in manufacturing, and it passes verification, then the verification becomes irrelevant. For production units, the risk is moderate/high depending on the product. Lack of training increases the risk of defective product used on patients. Depending on the severity of patient harm, Pathway would need to release an MDR and/or recall of the impacted product.
Deviations: <i>(Were there any deviations from the Investigation Plan? If yes, provide justification. If no deviations, indicate "N/A")</i>	N/A
Root Cause: <i>(Describe the root cause(s) by performing the following and attaching any supporting documents. Attach any supporting documents. If needed, complete and attach the SCAR Form, FM-0087)</i> • Identify the problem • Gather the data • Analyze the problem • Solve the problem	Problem: Manufacturers do not have training records or have late training records 1. Man a. Lost physical copies of training records on engineering builds i. Individuals acquiring training files were not trained to Arena upload ii. No defined owner of training process iii. Engineers not trained to training requirements for production b. Individuals unaware of updated revisions released in Arena c. Failed to train before manufacturing initiated, and signed-off a training record post-build 2. Method a. Lack of protocol for training requirements for DV/Engineering builds b. Lack of agreement/protocol for non-Pathway employees' training requirements

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Use 5 WHY Analysis, Fishbone Diagram, Pareto Analysis, 6M (Man, Method, Material, Machine, Measurement, Mother Nature).	c. No implementations or training files linked in Arena for ECO/DEVs containing MP/WI updates d. Training files not required before initiating manufacturing i. No checks/stops in manufacturing process from start to finish ii. No system to review training status of all employees in manufacturing e. Retraining is not required for any current documents f. GDP and GMP is not explicitly trained
Root Cause Code(s): <i>(check all that apply)</i>	☒ Code 1 – Man – Inadequate Training ☐ Code 2 – Man – Procedure Not Used/Used Improperly/Step Omitted ☐ Code 3 – Man – Human Error ☐ Code 4 – Man – Lack of Qualification, Demonstrated Proficiency ☒ Code 5 – Method – Incomplete Procedure/Wrong/Ambiguous/Non-Existent ☐ Code 6 – Method – Manufacturing Process Controls ☐ Code 7 – Method – Validation/Verification Issue ☐ Code 8 – Material – Damaged/Contaminated ☐ Code 9 – Material – Invalid Testing ☐ Code 10 – Material – Testing not met ☐ Code 11 – Material – Expired/Damaged ☐ Code 12 – Material – Specification Incomplete/Wrong/Ambiguous ☐ Code 13 – Material – Device Biocompatibility ☐ Code 14 – Material – Material Compatibility ☐ Code 15 – Material – Visual Specification not met/Unusual characteristics ☐ Code 16 – Material – Incorrect Storage ☐ Code 17 – Machine – Defective Machine/Tool ☐ Code 18 – Machine – Machine/Tool Capacity ☐ Code 19 – Machine – Operational Failure ☐ Code 20 – Machine – Equipment Controls and Calibration ☐ Code 21 – Machine – Inadequate Maintenance ☐ Code 22 – Environmental Controls – Facilities/Utilities/Clean Room ☐ Code 23 – Environmental Controls – Temperature/Humidity ☐ Code 24 – Environmental Controls – Cleanliness/Particulate ☐ Code 25 – Environmental Controls – Microbial ☐ Code 26 – Other – Documentation Issue ☐ Code 27 – Other – Labeling Issue ☐ Code 28 – Other – Manufacturing Issue ☐ Code 29 – Other – Supplier Issue ☐ Code 30 – Other – Design Issue: _____
	☐ Attachments

Phase 3 Target Due Date:	04/25/2025
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PHASE 2 – APPROVAL SIGNATURES

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CAPA Owner (Name/Title):	Sean Tamayo / Pathway Manufacturing	Signature/Date:	 Sean Tamayo (27-Mar-2025 05:23 PDT)
Executive Management (Name/Title):	Aaron Rogers / Pathway Director of Regulatory and Quality	Signature/Date:	 Aaron Rogers (26-Mar-2025 13:59 PDT)
Quality Assurance (Name/Title):	Larry Dacoron / Pathway Senior Quality Engineer	Signature/Date:	 Larry Dacoron (01-Apr-2025 17:34 PDT)

CAPA EXTENSION REQUEST (☒ N/A if not applicable)

CAPA Extension Due Date:	N/A
Justification for Extension:	N/A

CAPA EXTENSION APPROVAL SIGNATURES (☒ N/A if not applicable)

CAPA Owner (Name/Title):	N/A	Signature/Date:	N/A
Quality Assurance (Name/Title):	N/A	Signature/Date:	N/A

CAPA-24-001 Phase 2

Final Audit Report

2025-04-01

Created:	2025-03-26 (Pacific Daylight Time)
By:	Jacqueline Kim (jkim@pathwaymedtech.com)
Status:	Approved
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Number of Documents:	1
Document page count:	6
Number of supporting files:	0
Supporting files page count:	0

"CAPA-24-001 Phase 2" History

-  Document created by Jacqueline Kim (jkim@pathwaymedtech.com)
2025-03-26 - 1:55:36 PM PDT
-  Document emailed to Sean Tamayo (stamayo@pathwaymedtech.com) for approval
2025-03-26 - 1:56:23 PM PDT
-  Document emailed to Aaron Rogers (arogers@pathwaymedtech.com) for approval
2025-03-26 - 1:56:23 PM PDT
-  Document emailed to Larry Dacoron (ldacoron@pathwaymedtech.com) for approval
2025-03-26 - 1:56:23 PM PDT
-  Email viewed by Aaron Rogers (arogers@pathwaymedtech.com)
2025-03-26 - 1:58:17 PM PDT
-  Aaron Rogers (arogers@pathwaymedtech.com) authenticated with Email OTP by verifying one-time code sent via email
2025-03-26 - 1:58:35 PM PDT
-  Agreement viewed by Aaron Rogers (arogers@pathwaymedtech.com)
2025-03-26 - 1:58:35 PM PDT
-  Aaron Rogers (arogers@pathwaymedtech.com) has agreed to the terms of use and to do business electronically with Pathway Medtech LLC
2025-03-26 - 1:59:18 PM PDT



Document approved by Aaron Rogers (arogers@pathwaymedtech.com)

Approval Date: 2025-03-26 - 1:59:18 PM PDT - Time Source: server



Email viewed by Sean Tamayo (stamayo@pathwaymedtech.com)

2025-03-27 - 5:22:30 AM PDT



Sean Tamayo (stamayo@pathwaymedtech.com) authenticated with Email OTP by verifying one-time code sent via email

2025-03-27 - 5:22:50 AM PDT



Agreement viewed by Sean Tamayo (stamayo@pathwaymedtech.com)

2025-03-27 - 5:22:52 AM PDT



Sean Tamayo (stamayo@pathwaymedtech.com) has agreed to the terms of use and to do business electronically with Pathway Medtech LLC

2025-03-27 - 5:23:31 AM PDT



Document approved by Sean Tamayo (stamayo@pathwaymedtech.com)

Approval Date: 2025-03-27 - 5:23:31 AM PDT - Time Source: server



Larry Dacoron (ldacoron@pathwaymedtech.com) authenticated with Email OTP by verifying one-time code sent via email

2025-04-01 - 5:34:05 PM PDT



Agreement viewed by Larry Dacoron (ldacoron@pathwaymedtech.com)

2025-04-01 - 5:34:05 PM PDT



Larry Dacoron (ldacoron@pathwaymedtech.com) has agreed to the terms of use and to do business electronically with Pathway Medtech LLC

2025-04-01 - 5:34:57 PM PDT



Document approved by Larry Dacoron (ldacoron@pathwaymedtech.com)

Approval Date: 2025-04-01 - 5:34:57 PM PDT - Time Source: server



Agreement completed.

2025-04-01 - 5:34:57 PM PDT